Bounded-memory streaming redaction of personal data and credentials with a whole-document equivalence guarantee

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Abstract

Logs, database exports and packet captures that leave an organisation carry personal data and credentials, and they are often too large to hold in memory. Processing them in chunks adds a silent failure: a value that straddles a chunk boundary reaches the detector in two halves, matches nothing, and is written out in the clear. We present SafeStream-Redactor, a local redaction engine that treats chunking as a correctness problem. Each detector declares the longest span it emits and the context it reads. If the window overlap covers every detector's span plus its right context, we prove that the streamed output equals a single whole-document pass, that multi-process output equals the sequential output or falls back to it, and that the working buffer is independent of input size. The engine derives the overlap from the declarations and enforces it at run time. Peak memory added by streaming stayed between 8.4 and 8.54 MiB from 8 MiB to 2 GiB of input, against 904.6 MiB for whole-document redaction of a 32 MiB file; throughput was 10.5 MB/s with one worker and 41.1 MB/s with four. On held-out test data the pattern tiers fully redacted 0.437 of credentials, against 0.311 for gitleaks and 0.270 for TruffleHog in default configurations. The guarantee does not make detection complete: no configuration fully protected more than 17.9% of RedactionBench documents, and two of the three corpora are synthetic.

Keywords: Data redaction, Secret detection, Data loss prevention, Streaming algorithms, Bounded memory, Personally identifiable information

1 Introduction

Text artefacts that leave an organisation, such as an application log attached to a support ticket, a database export handed to an analyst, or a packet capture shared with a vendor, routinely carry two classes of sensitive value: personal data (e-mail addresses, telephone and card numbers, national identifiers, names) and credentials (application programming interface keys, access tokens, private keys, passwords). Data-loss-prevention systems inspect such content before it crosses a boundary [1], and measurement studies show that credentials leak into shared artefacts at scale [2, 3].

Two families of tools address the problem. Personal-data engines such as Microsoft Presidio [4] combine pattern recognisers with a statistical named-entity model but ship no credential recognisers. Secret scanners such as gitleaks [5] and TruffleHog [6] target credentials in source repositories, and their accuracy varies widely across secret types [7]. The artefacts that most need redaction are often too large to analyse in one call, so they are processed in pieces. A chunked redactor can then fail silently: a value that straddles a chunk boundary is presented to the detector in two halves, neither of which matches, and both halves are written out in the clear. The same problem is long established in network security, where signatures split across packets motivated stream reassembly in intrusion detection [8] and streaming modes in regular-expression engines [9, 10]. In redaction, however, detectors read context around a match ("Patient ID:" before a number, a key name before a token), confidences are adjusted by neighbouring text, and overlapping detections are merged before the text is rewritten. Overlapping windows are the obvious remedy, but how much overlap is enough depends on all of these, and none of the redaction or secret-scanning systems we examined states when its chunked output equals its whole-document output.

This paper studies that question for a concrete system, SafeStream-Redactor (hereafter SafeStream). The argument has four parts.

1. **Equivalence under declared bounds.** Every detector d declares an upper bound L_d on the span it can emit and bounds $C_{\text{left},d}$ and $C_{\text{right},d}$ on the context it reads (Assumptions A1 and A2). If the overlap w is at least $\max_d \max(L_d + C_{\text{right},d}, C_{\text{left},d})$, a streaming engine produces exactly the whole-document result, and a multi-process engine with a seam cross-check produces exactly the sequential result or falls back to it (Section 5, Appendix A). The argument is a sliding-window locality argument; its contribution is the instantiation for context-dependent redaction, with constants derived from the detectors and enforced at run time rather than chosen by hand. For the default pipeline the requirement is exactly the default overlap of 8192 characters.
2. **Bounded memory.** The working buffer is bounded independently of input size, and measured peak memory stays flat across a 256-fold range of input sizes (Section 7.2).
3. **Detection quality, measured separately.** Equivalence says nothing about how good the detectors are. We evaluate them on held-out test splits against Presidio and two secret scanners, with paired bootstrap intervals at the document, cluster and two-stage (hierarchical) levels, an ablation and a contamination analysis (Sections 6 and 7).

4. **Where it stops.** Document-level protection is weak, the corpora are mostly synthetic, and an optional neural tier was not evaluated; we state the conditions under which the approach is useful and what evidence a deployment would need (Section 8).

[MISSING: If the FTNCT'09 conference paper on SafeStream-Redactor was published, cite it here and state what this article adds: held-out test evaluation, secret-scanner baselines, cluster-robust statistics, ablation, contamination analysis, peak-memory curve, benign-shape suppression in the entropy tier, native core parity results.]

Two scope limits are stated at the outset. First, the guarantee concerns equivalence rather than accuracy: streaming does not improve detection, it makes the streamed result the same as the whole-document result, so every detection error of the whole-document pipeline is also an error of the streamed one. Second, the evaluation has known weaknesses that we quantify rather than omit: one corpus, RedactionBench, was inspected during rule development (Section 7.5); two corpora are synthetic; and fewer than one RedactionBench document in five was fully protected by any configuration we evaluated (Section 7.5).

2 Background and related work

Personal-data detection and de-identification

Rule-based de-identification of clinical text dates back to lookup tables and regular expressions applied to nursing notes [11]; Philter refined the approach on a large annotated clinical corpus [12]. The 2014 i2b2/UTHealth shared task established annotated corpora and evaluation practice [13], and recurrent neural networks reached strong results on them [14]. Presidio [4] combines pattern recognisers with spaCy or transformer models and ships an evaluation toolkit with a synthetic dataset [15]. Lison et al. argue that sequence labelling alone does not amount to anonymisation and call for explicit disclosure-risk measures [16]; the Text Anonymization Benchmark operationalises this for court judgments [17]. Named-entity recognition underpins the free-text part of these systems, from the CoNLL-2003 shared task [18] to transformer encoders [19, 20] and generalist span models [21]. Pseudonymisation with a keyed hash such as HMAC [22] preserves referential integrity but reveals equality between values, so it is not anonymisation in the sense of k-anonymity [23]. Redaction as studied here is narrower than anonymisation: it removes spans that detectors recognise and makes no claim about re-identification from what remains.

Credential and secret detection

Meli et al. measured secret leakage across public GitHub repositories [2]. Approaches range from vendor-specific regular expressions [24] to machine-learned filters that reduce false positives [25] and neural detectors for free-form passwords [26]. SecretBench provides a labelled corpus of candidate secrets [27], and a comparison of nine tools attributed false positives to generic patterns and entropy heuristics and false negatives to missing rules [7]. Krause et al. studied how developers handle leaked secrets [3]. Shannon entropy [28] is the usual signal for undocumented high-entropy secrets. The Nosey Parker rule set [29] supplies the labelled examples we evaluate on.

Detection across segment boundaries

That a pattern split across units of input can evade a detector is well known in network security: Ptacek and Newsham showed that intrusion detectors which do not reconstruct streams as the end host does can be evaded [8], and Hyperscan [9] provides a streaming mode whose matches do not depend on how the data is split into blocks [10]. These systems carry automaton state across boundaries for pattern matching. A redaction pipeline differs in three ways that affect correctness: detectors consult bounded context on both sides of a match, confidences depend on nearby text, and overlapping detections are resolved jointly before the output is rewritten. Overlapping windows are a common way to process long inputs, and the stream model [30] formalises computation in bounded memory; linear-time regular-expression matching goes back to Thompson [31]. Among the redaction, de-identification and secret-scanning systems cited above, we found none that states when chunked output equals whole-document output or how the overlap must be sized. We do not claim that the underlying locality argument is new; the contribution is its instantiation and enforcement for context-dependent redaction.

3 Problem statement and threat model

Definition 1 (Detections and redaction) A text T is a finite character string, and $T[a{:}b]$ is its half-open substring. A *detection* is a tuple (i, j, τ, c) with $0 \le i < j \le |T|$: a half-open span, an entity type and a confidence in $[0, 1]$. Each detection is produced by one detector d of a finite set $\mathcal{D}$. A *pipeline* Π maps T to a set of pairwise-disjoint detections. *Redaction* rewrites T by replacing each span in $\Pi(T)$ according to a policy (replace, mask, hash, keyed pseudonym or encryption). The replacement of a span depends only on its type and its text, so redaction is a deterministic function of T and $\Pi(T)$ that acts on each span separately.

Definition 2 (Leak) Given a set G of gold sensitive spans for T, a *leak* is any character of a span in G that is not covered by a span in $\Pi(T)$. A gold span with at least one uncovered character is *leaked*, because a partly visible card number or key can often be reconstructed or narrowed.

Adversary and setting

The adversary is a later reader of the redacted output, for example the recipient of a shared log. The input may be larger than memory and arrive as an arbitrary sequence of chunks. The system runs locally and makes no network calls, so the unredacted text does not leave the host.

What is in scope

(i) That chunked and parallel processing introduce no leak beyond those of the whole-document pipeline; (ii) bounded working memory; (iii) the detection quality of the whole-document pipeline, measured empirically.

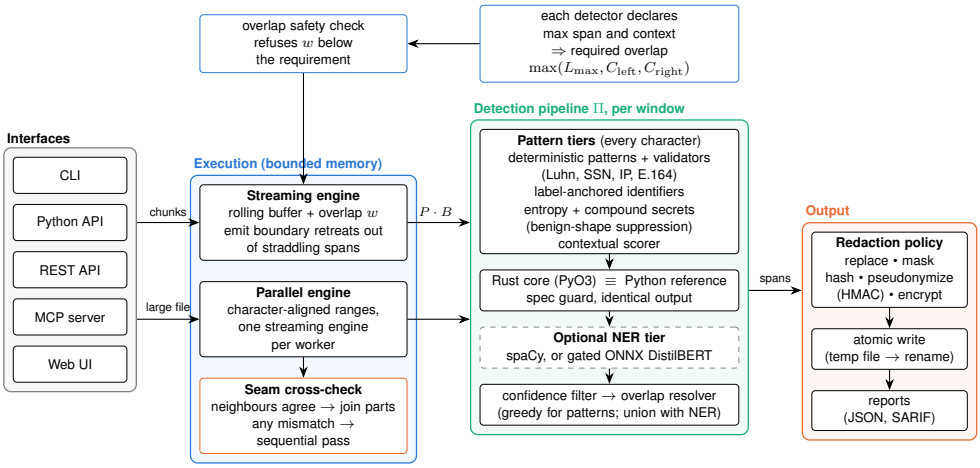


Fig. 1 System architecture. Interfaces feed text to the streaming engine or, for a large file, to the parallel engine, whose workers each run a streaming engine and whose seams are cross-checked. Each window passes through the detection pipeline Π: the pattern tiers run on every character, on the Rust core or the identical Python reference. The dashed named-entity tier is optional; its spaCy configuration was evaluated, its ONNX configuration was not. The resolver produces disjoint spans, which the redaction policy rewrites and an atomic write commits. The overlap w is checked against the span and context bounds the active detectors declare.

What is out of scope

Detection completeness: the pipeline misses values its detectors do not recognise, and the guarantee does not change that. Adversarial evasion: an author who encodes, splits or obfuscates a secret on purpose is not modelled. Inference from residual text: quasi-identifiers and context that remain after redaction may still allow re-identification [16, 23]. Side channels such as output length. Bounded memory is a resource property, not a security property.

4 Design

Figure 1 shows the architecture. The reference pipeline Π runs every configured detector on a text, adjusts confidences with a bounded contextual scorer, drops detections below a threshold (0.5 by default) and resolves overlaps into a disjoint set.

4.1 Detection tiers

Deterministic patterns with validators

Credential formats with a fixed shape (cloud access keys, provider tokens, JSON Web Tokens, PEM private-key blocks) and structured personal data (e-mail, telephone, card and social-security numbers, IPv4 and IPv6 addresses) are matched by regular expressions. A match becomes a detection only if it passes a validator: the Luhn checksum for card numbers [32], area, group and serial rules for social-security numbers,

octet ranges for IPv4, address parsing for IPv6, and an E.164 shape check for telephone numbers. A generic key-assignment rule matches values that follow keys such as `api_key`, `token` or `password`.

Label-anchored identifiers

Unchecked numeric identifiers such as medical record, account or licence numbers are ambiguous in isolation, so this tier promotes a value to a detection only when a cue word that names it (for example `Patient ID:`) appears within 96 characters to its left.

Entropy tier with benign-shape suppression

Unstructured candidate runs of 20 to 8192 characters over a key alphabet are scored by length, character-class mix and Shannon entropy [28], with a lower threshold for hexadecimal runs; a compound variant covers runs with embedded symbols. Entropy alone cannot separate a secret from other random-looking strings. During this study every generated UUID, commit hash, SHA-256 digest and base64 image payload in the secrets development split was flagged (Section 7.4). The tier therefore suppresses three shapes: a canonical UUID, a 40- or 64-character hexadecimal value preceded by a digest cue (`commit`, `sha256`, `hash` and similar), and the payload of a `data:` URI. Suppression is vetoed when a secret-like key (`key`, `token`, `secret`, `password` and similar) names the value, so `SAUCE_ACCESS_KEY=`*uuid* remains a secret. The check reads at most 32 characters of left context.

Contextual scorer

A 48-character left window raises or lowers confidences when trigger words are present and emits a small set of trigger-gated detections of its own.

Optional named-entity tier

For names, organisations and locations, an optional tier runs a spaCy pipeline (`en_core_web_sm` in our experiments). A transformer tier with a regular-expression gate is also implemented but was not evaluated and contributes no result to this paper (Section 8.4).

4.2 Overlap resolution

Pattern detections that overlap are resolved best-first by confidence, then by length, so a card number beats a telephone number embedded in it. Named-entity detections are handled differently: any cluster of overlapping spans that contains a named-entity span becomes one span from the cluster's leftmost start to its rightmost end, labelled by its best pattern member if one exists. Without this union rule, a PERSON tag on `john.doe` could replace the e-mail detection for `john.doe@example.com` and leave the domain exposed.

Algorithm 1 Streaming detection with overlap w

Require: chunks $c_1, c_2, \ldots$; pipeline Π; overlap w; minimum emit size $m \geq w$

$P \leftarrow \varepsilon$ ▷ emitted text kept as read-only left context
$B \leftarrow \varepsilon$ ▷ buffer of unemitted text
for each chunk c **do**
  $B \leftarrow B \cdot c$
  if $|B| < w + m$ **then**
    continue
  end if
  $S \leftarrow \{(i - |P|,\ j - |P|,\ \tau,\ \kappa) : (i, j, \tau, \kappa) \in \Pi(P \cdot B),\ i \geq |P|\}$
  $e \leftarrow |B| - w$ ▷ planned emit boundary
  while some $(i, j, \tau, \kappa) \in S$ has $i < e < j$ **do**
    $e \leftarrow i$ ▷ retreat out of the straddler
  end while
  if $e \leq 0$ **then**
    continue ▷ a span pins the buffer; read more
  end if
  emit $B[0{:}e]$ with $\{(i, j, \tau, \kappa) \in S : j \leq e\}$
  $P \leftarrow$ last w characters of $P \cdot B[0{:}e]$
  $B \leftarrow B[e{:}]$
end for
emit B with $\{(i - |P|,\ j - |P|,\ \tau,\ \kappa) : (i, j, \tau, \kappa) \in \Pi(P \cdot B),\ i \geq |P|\}$

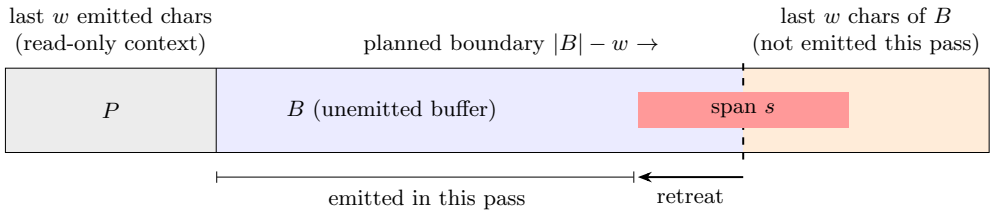


Fig. 2 One pass of Algorithm 1. The detector reads $P \cdot B$; only text before the planned boundary may be emitted, and the boundary retreats out of any span that crosses it, so s is carried to the next pass and detected whole.

4.3 Streaming engine

Algorithm 1, a simplified form of `engine.stream_windows` illustrated in Figure 2, keeps a buffer of unemitted text and the last w characters already emitted. Each pass detects over their concatenation, so detectors see real left context rather than an artificial edge, and emits only the part of the buffer that lies at least w characters before its end. If a detection crosses the planned emit boundary, the boundary retreats to the detection's start and the detection is carried into the next pass. Small chunks are coalesced until at least m characters can be emitted, which bounds the cost of

rescanning the overlap per emitted character regardless of how the input is chunked. The defaults are $w = 8192$ characters and $m = 65{,}536$ characters.

4.4 Parallel engine

For large files the parallel engine splits the input into character-aligned byte ranges of at least $4w$ characters, one per worker process. Each worker reads w characters of scanned padding and a further w characters of context-only prefix around its interior boundaries (seams). A span that crosses a seam belongs to the left worker, and the right worker begins emitting at that span's end. Neighbouring workers compute the seam position independently, and the coordinator compares their reports; on any disagreement it discards the parallel result and redacts the file sequentially.

4.5 Implementation

The pure-Python implementation is the reference. An optional Rust core implements the pattern tiers. At start-up it checks that its compiled pattern sources equal the Python ones, and it hands a window back to Python in the rare cases where Unicode case folding or floating-point summation could make the two disagree.

Use of AI assistance

An AI assistant (Claude, Anthropic) was used to implement evaluation scripts and to draft this manuscript under the authors' supervision. The authors reviewed every result and every sentence and are accountable for the final content.

5 Formal analysis

Let $S_w(T)$ be the set of detections emitted by Algorithm 1 for some chunking of T, mapped back to absolute offsets, and $P_{w,k}(T)$ the output of the parallel engine with k workers. Each detector $d \in \mathcal{D}$ declares three constants: the longest span it emits, L_d, and the context it reads to the left and right of a span, $C_{\text{left},d}$ and $C_{\text{right},d}$. The two assumptions are stated per detector and must hold for every input text, not only for T, because the engine applies Π to windows.

Assumption A1 (Bounded span) For every text X, every detection of detector d in $\Pi(X)$ has length $j - i \le L_d$.

Assumption A2 (Bounded context) Let X and Y be texts and δ an offset with $X[a{:}b] = Y[a{+}\delta : b{+}\delta]$. Let (i, j) be a span of detector d such that either $i - C_{\text{left},d} \ge a$ or $a = a + \delta = 0$, and either $j + C_{\text{right},d} \le b$ or both $b = |X|$ and $b + \delta = |Y|$. Then $(i, j, \tau, \kappa) \in \Pi(X)$ if and only if $(i + \delta, j + \delta, \tau, \kappa) \in \Pi(Y)$.

In words, A2 says that whether a span of d is in the final detection set, and with which type and confidence, depends only on the $C_{\text{left},d}$ characters before it, the span

itself and the $C_{\text{right},d}$ characters after it, where a window edge that coincides with an edge of both texts counts as real. The pipeline requirement is

$$w^\star \;=\; \max_{d\in\mathcal{D}} \max\big(L_d + C_{\text{right},d},\; C_{\text{left},d}\big). \tag{1}$$

Theorem 1 (Streaming equivalence) *Under A1 and A2, if $w \geq w^\star$ and $m \geq w$, then $S_w(T) = \Pi(T)$ for every text T and every chunking of T.*

Corollary 1 (Pipeline-wide sufficient condition) *Let $L_{\max} = \max_d L_d$, $C_{\text{left}} = \max_d C_{\text{left},d}$ and $C_{\text{right}} = \max_d C_{\text{right},d}$. Then $w \geq \max(L_{\max} + C_{\text{right}}, C_{\text{left}})$ implies $w \geq w^\star$, so Theorem 1 applies.*

The corollary is the intuition: the overlap must hold the longest span together with the widest right context, and the widest left context. It is sufficient but not necessary, because it pairs the longest span with a right context that may belong to a different detector. Equation (1) is the condition the engine checks (Section 5.1).

Proof of Theorem 1 Fix a non-final pass. Let p be the absolute position of $B[0]$, $q = p + |B|$ and $e_0 = q - w$ the planned boundary. The window is $W = T[p - |P| : q]$, where $|P| = \min(w, p)$ because P holds the last w emitted characters.

Safe region. Let $s = (i, j, \tau, \kappa)$ be a span of detector d, in absolute offsets, with $p \leq i < e_0$ that belongs to $\Pi(W)$ or to $\Pi(T)$. By A1, $j \leq i + L_d < q - w + L_d \leq q - C_{\text{right},d}$, using $w \geq L_d + C_{\text{right},d}$. On the left, $i - C_{\text{left},d} \geq p - w$ because $w \geq C_{\text{left},d}$; if $p < w$, the window starts at position 0 of T. A2, applied to $X = W$ and $Y = T$, then gives $s \in \Pi(W) \Leftrightarrow s \in \Pi(T)$. So the spans of the window that start in $[p, e_0)$ are exactly the spans of $\Pi(T)$ that start there.

Retreat. $\Pi(W)$ is pairwise disjoint, so at most one span s^* straddles e_0, and it starts before e_0, so it belongs to $\Pi(T)$. The boundary becomes $e = i^*$ if s^* exists and $e = e_0$ otherwise. In the first case, $i^* > e_0 - L_d \geq p + m - L_d \geq p$, because $|B| \geq w + m$ and $m \geq w \geq L_d$. So $e > p$, and the pass emits at least one character; the pinned branch is never taken. No span of $\Pi(T)$ straddles e. If $e = e_0$, such a span would start before e_0 and so be the straddler of W. If $e = i^*$, it would overlap s^*, which disjointness rules out.

Emission. The pass emits $T[p{:}e]$ with the window spans contained in it. By the safe region these are exactly the spans of $\Pi(T)$ contained in $[p, e)$. The final pass has $q = |T|$, so the right-hand clause of A2 holds for every span, and it emits exactly the spans of $\Pi(T)$ inside $[p, |T|)$. The emitted segments partition T, and no span of $\Pi(T)$ crosses a segment boundary. Every span of $\Pi(T)$ is therefore emitted exactly once, with the type and confidence it has in $\Pi(T)$, and nothing else is emitted. □

Theorem 2 (Parallel identity with fail-safe fallback) *(i) Under the conditions of Theorem 1, and with ranges of at least $4w$ characters, every seam report agrees and $P_{w,k}(T)$ equals the sequential output byte for byte. (ii) Without any assumption on the detectors, the output is either assembled from parts whose seam reports all agree, so that every character of T is emitted exactly once, or it is the sequential output.*

Theorem 3 (Bounded working buffer) *Let c be the length of the longest input chunk. Under the conditions of Theorem 1, the unemitted buffer of S_w never exceeds $w + m + c$ characters, and each detection window never exceeds $2w + m + c$, independent of $|T|$. Each parallel worker*

Table 1 Declared streaming bounds of the evaluated detectors, in characters, at the evaluated commit. The requirement of a detector is $\max(L_d + C_{\text{right},d},\, C_{\text{left},d})$; the pipeline requirement $w^\star$ is the largest requirement among active detectors.

Detector d	L_d	$C_{\text{left},d}$	$C_{\text{right},d}$	Requirement
Deterministic patterns	8192	0	0	8192
Label-anchored identifiers	64	96	8	96
Entropy tier	8192	32	0	8192
Compound entropy tier	8192	0	0	8192
Contextual scorer	9	48	0	48
spaCy named-entity tier (optional)	256	256	256	512
Default pipeline, $w^\star$				8192
Default pipeline with spaCy, $w^\star$				8192

additionally holds back at most $2w$ *characters plus one segment, so the parallel engine needs* $O\big(k\,(w + m + c)\big)$ *characters.*

Because redaction acts on each span separately (Definition 1), Theorem 1 implies byte-identical redacted output. The proofs of Theorems 2 and 3, and a table mapping each step to the implementation, are in Appendix A. For `redact_file`, $c = m =$ 65,536 and $w = 8192$, so the buffer stays below 139,264 characters and each window below 147,456 characters. The proofs were not machine-checked.

5.1 What is derived and what is assumed

Each detector declares $(L_d, C_{\text{left},d}, C_{\text{right},d})$. The engine computes $w^\star$ from Eq. (1) over the active detectors and refuses to start with a smaller overlap unless the caller explicitly overrides it. Table 1 lists the declarations. The detectors that emit the longest spans (private-key blocks and entropy candidates, $L_d = 8192$) read no right context. The only detector with right context among the default tiers, the label-anchored tier, emits spans of at most 64 characters. Hence $w^\star = 8192$, and the default overlap $w = 8192$ satisfies Theorem 1 exactly, with or without the spaCy tier. The pipeline-wide Corollary 1 would ask for $8192 + 8 = 8200$ characters; that is why the per-detector form is the one the engine enforces. A regression test fails if any declaration changes, and a differential test probes each detector to check that its real context dependency does not exceed its declaration.

Two parts of A1 and A2 are *not* derived from declarations. They should be read as assumptions corroborated by tests rather than proven properties of the code. (i) The confidence adjustment and threshold must not reach beyond the declared context of the detector whose span they adjust. The contextual scorer's 48-character left window is within w for every detector, but it is not added to the other detectors' declarations. (ii) Overlap resolution must not propagate decisions along chains of overlapping candidates beyond the declared contexts, and the named-entity union rule must not create spans longer than the declared bounds. For the pattern tiers, whose spans are bounded and whose overlaps are short, the equivalence tests in Section 7.1 found no counterexample. A statistical model's output can depend on its whole input window, so for the spaCy tier the declared bounds of 256 characters are declarations only, and the theorems should not be read as covering it.

Failure modes

Lowering the overlap below $w^\star$ is refused by default. If a caller overrides that, the parallel engine still cannot lose or duplicate characters (Theorem 2(ii)), but the sequential engine can split a long span. A detector that understates its context, or one that needs document-wide context, is outside the guarantee.

6 Experimental methodology

The evaluation addresses five questions. **RQ1**: do the implementation's streamed, parallel and native outputs match the reference in practice? **RQ2**: how do memory and throughput scale? **RQ3**: how well do the pattern tiers detect credentials compared with purpose-built scanners? **RQ4**: how well does SafeStream detect personal data compared with Presidio, with and without a named-entity model? **RQ5**: how much protection does it give at the level of a whole document, and how much of that depends on rules written after seeing the benchmark?

6.1 Corpora and splits

Table 2 lists the test splits. Each corpus was split once into development and test halves with a fixed seed. The manifests list document identifiers and the SHA-256 of the source data, and the loader refuses a source that no longer matches. Development, error analysis and tuning used development halves only, with the exception of RedactionBench described below; test halves were scored after the code was frozen, with one disclosed exception (Section 7.4).

- **presidio-research** [15] (synthetic): the dataset distributed with Presidio's evaluation toolkit (revision 0cb3650), with person, location, organisation, telephone, card, date and other labels, generated from templates. The 750 test documents come from 200 templates (median 3.5 documents per template).
- **secrets** (semi-synthetic): the labelled examples in the Nosey Parker rule set [29] (revision 2e6e7f3) as positives, with gold spans given by each rule's own capture groups, plus 200 generated non-secrets (UUIDs, commit hashes, SHA-256 digests, base64 image payloads, minified JavaScript) as gold-free documents. The split is by rule, so the test half contains vendor formats absent from the development half; its 216 positives come from 96 rules. The positives are short examples written by rule authors, not credentials observed in real artefacts.
- **RedactionBench** [33]: 200 documents across eleven domains, a majority seeded from real-world sources according to its authors, with character spans labelled *mandatory* or *contextual*. The test half holds between 5 and 17 documents per domain. All 200 documents were used for error analysis and rule writing before the split existed, so its test half is not held out (Section 7.5).

6.2 Systems and configurations

SafeStream was run with the Rust core and default tiers (*pattern*), with the opt-in date tier (*+dates*), and with the spaCy `en_core_web_sm` tier (*+spaCy*). **Presidio**

Table 2 Test splits. Gold-free documents contain no gold span; any detection on them is a false alarm. Clusters are the resampling units of the cluster and two-stage bootstrap (Section 6.4).

Corpus	Documents	Gold spans	Gold-free	Bytes	Clusters
presidio-research	750	1,394	63	63,678	200 templates
secrets	313	293	97	46,595	96 rules + 5 templates
RedactionBench	100	25,592	0	1,378,267	11 domains

2.2.355 [4] was run as its pattern recognisers alone (*regex*) and as the full analyser with spaCy `en_core_web_sm` (*sm*) and `en_core_web_lg` (*lg*), with spaCy 3.8.13 and models 3.8.0. **gitleaks** 8.28.0 [5] and **TruffleHog** 3.90.8 [6] were run with default rules and default false-positive filtering; TruffleHog ran with `--no-verification`, so no example credential was sent to a live service. Each scanner ran once over a directory with one file per document, and findings were mapped back to character offsets; TruffleHog's composite findings (for example `user:token`) were credited for each component that appears in the text. Each Python tool ran in its own virtual environment and process, pinned to one thread; the scanners are Go binaries and ran with their default concurrency. All SafeStream thresholds are defaults; no threshold was tuned on test data. The RedactionBench dev-selected configurations (Section 7.5) were chosen on the development half by a rule fixed in advance.

6.3 Metrics

Let G and R be the sets of gold and redacted character positions in a document. Character precision and recall are $|G \cap R|/|R|$ and $|G \cap R|/|G|$, summed over documents, and F1 is their harmonic mean. All metrics are type-agnostic: a value counts as protected whatever label it receives. The *fully redacted rate* is the share of gold spans with every character redacted. The *false-alarm document rate* is the share of gold-free documents with at least one detection. On RedactionBench, *mandatory recall* is character recall over mandatory spans, and the *no-leak document rate* is the share of documents in which every mandatory character is redacted (Section 7.5 discusses how it relates to recall).

6.4 Statistical analysis

We report 95% percentile bootstrap intervals with 2,000 resamples and seed 0 [34]. All systems evaluated on a corpus are scored on one shared resample matrix, so every difference between two systems is paired; we report the median paired difference and call it significant only when its paired interval excludes zero. Documents are not independent. Secrets positives cluster by rule, presidio-research documents share generator templates, and RedactionBench documents share domains. We therefore use three resampling schemes on the same per-document counts [35]: *document* (documents drawn independently), *cluster* (whole clusters drawn with replacement, all their documents kept) and *two-stage* (clusters drawn with replacement, then documents drawn with replacement inside each drawn cluster). Table 2 lists the clusters. Tables 4, 5 and 7 give document-level intervals. Table 6 repeats the planned comparisons under all three schemes, and we call a difference *robust* when it is significant under all three.

RedactionBench has only eleven clusters, so its cluster intervals are themselves imprecise; we read them as a sensitivity analysis rather than as an exact test. No correction for multiple comparisons was applied; the comparisons in the abstract were planned, the ablation comparisons were not. The analysis was not pre-registered.

6.5 Environment

Accuracy runs used a 4-vCPU Intel Xeon virtual machine at 2.80 GHz with 15 GB RAM, Linux and CPython 3.13. Throughput figures come from separate runs, five per configuration, on the same machine with no other benchmark running; the throughput recorded during accuracy runs is not used. Memory was measured as the peak resident set size of a fresh process per point minus its resident set size after imports, so native allocations are included; it is reported in MiB. The cluster analysis re-ran every test-split system at the evaluated commits with the same tool versions and saved per-document predictions; each point estimate and each document-level paired interval reproduced the published values to three decimals.

7 Results

7.1 RQ1: equivalence and backend parity

On every test split the Rust core and the pure-Python reference produced identical predictions (presidio-research 178 spans, secrets 298, RedactionBench 3,167, from the frozen commit). The repository's equivalence suite asserts byte-identical streamed, whole-document and parallel output over exhaustive small chunkings, property-based generated inputs, seam-straddling entities and forced fallbacks, and also checks native parity and declared context; it passed 133 of 133 tests at the final commit. These tests corroborate the theorems for the configurations they exercise; they are not proofs, and they do not exercise the named-entity tier's full input dependence.

7.2 RQ2: memory and throughput

Figure 3 plots peak resident memory against input size. Each point is a fresh process redacting a generated log file dense with personal data and credentials. With the Rust core, streaming `redact_file` added between 8.4 and 8.54 MiB at every size from 8 MiB to 2 GiB, a 256-fold range. The pure-Python backend added between 4.3 and 4.4 MiB from 8 to 512 MiB; it was not run at larger sizes because of its run time (726 s at 512 MiB). Whole-document `redact()` on the same generated inputs, measured the same way, added 235.5 MiB at 8 MiB and 904.6 MiB at 32 MiB, about 28 times the input size, because the input string, the detections and the output are held at once; at 32 MiB the two modes therefore differ by a factor of about 106 on identical input. A 128 MiB whole-document run was stopped after 20 minutes without completing and is not reported. These measurements are consistent with Theorem 3 but cover one generated input shape on one machine, and only two whole-document sizes.

Table 3 reports throughput on the test splits. Per-document overhead dominates on presidio-research and secrets, whose documents average 85 and 149 bytes; RedactionBench's documents average 13.8 kB. There the Rust core processed 10.4 ±

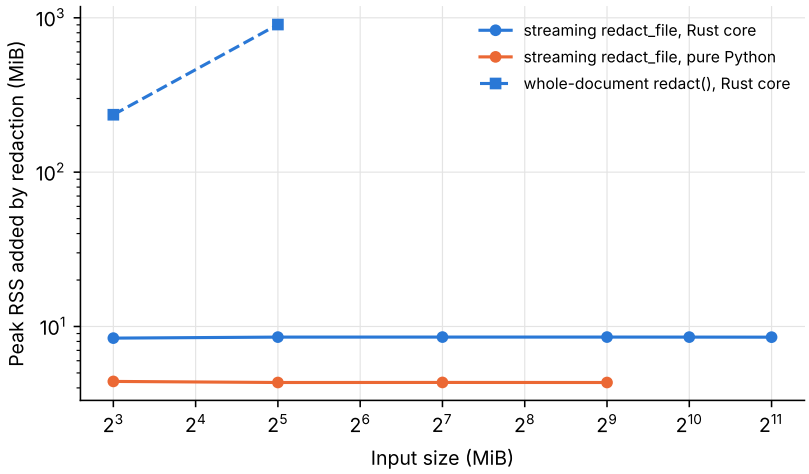


Fig. 3 Peak resident memory added by redaction (peak RSS minus RSS after imports, MiB) against input size, log scales, each point a fresh process. Streaming `redact_file` stays flat from 8 MiB to 2 GiB (Rust core) and to 512 MiB (pure Python); whole-document `redact()` grows with the input.

Table 3 Throughput on the test splits in MB/s, mean ± sample standard deviation over 5 runs, one thread, load time excluded. Each run is a fresh process on an otherwise idle 4-vCPU machine.

System	presidio-research	secrets	RedactionBench
SafeStream pattern (Rust core)	1.82 ± 0.42	2.39 ± 0.58	10.4 ± 1.3
SafeStream pattern (pure Python)	1.10 ± 0.07	0.916 ± 0.17	1.23 ± 0.15
SafeStream +spaCy (sm)	0.0118 ± 0.0009	0.0227 ± 0.0034	0.0467 ± 0.0033
Presidio regex	0.0802 ± 0.0127	0.0983 ± 0.0155	0.0694 ± 0.0028
Presidio sm	0.0081 ± 0.0004	0.0125 ± 0.0007	0.0201 ± 0.0003
Presidio lg	0.0074 ± 0.0010	0.0128 ± 0.0008	0.0203 ± 0.0003
gitleaks[1]	0.507 ± 0.028	0.409 ± 0.051	3.15 ± 0.19
TruffleHog[1]	0.0269 ± 0.0016	0.0188 ± 0.0007	0.594 ± 0.020

[1]Go binaries run once per corpus over a directory of documents; the figure includes process start-up and is not limited to one thread.

1.3 MB/s, the pure-Python reference 1.23±0.15 MB/s and Presidio's pattern recognisers 0.069 MB/s. With a statistical model in the loop every system ran below 0.05 MB/s: SafeStream +spaCy at 0.047 MB/s and Presidio sm and lg at 0.020 MB/s. The scanner figures include process start-up and are not comparable per thread.

On a single 256 MB generated log file (mostly benign log lines and prose with personal data and credentials mixed in), `redact_file` with the Rust core ran at 10.5 ± 0.5, 20.6 ± 0.5 and 41.1 ± 1.4 MB/s with 1, 2 and 4 worker processes, a 3.9-fold speed-up on 4 vCPUs (Figure 4). The pure-Python backend on a 32 MB file of the same kind ran at 0.99, 1.90 and 2.67 MB/s. These figures come from one virtual machine and should be read as relative rather than absolute.

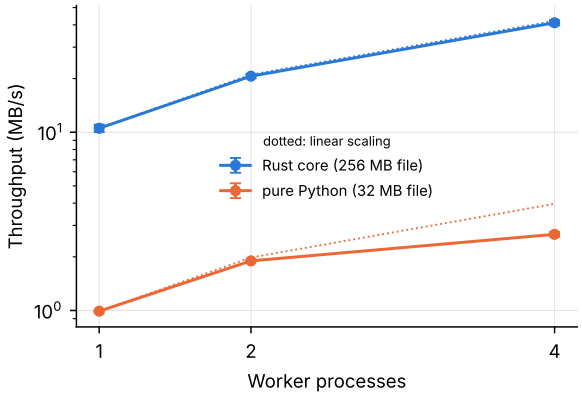


Fig. 4 Throughput of `redact_file` against worker processes on a 4-vCPU machine, mean and standard deviation over 5 runs (log scale). Dotted lines show linear scaling from one worker. The parallel output is byte-identical to the sequential output (Theorem 2).

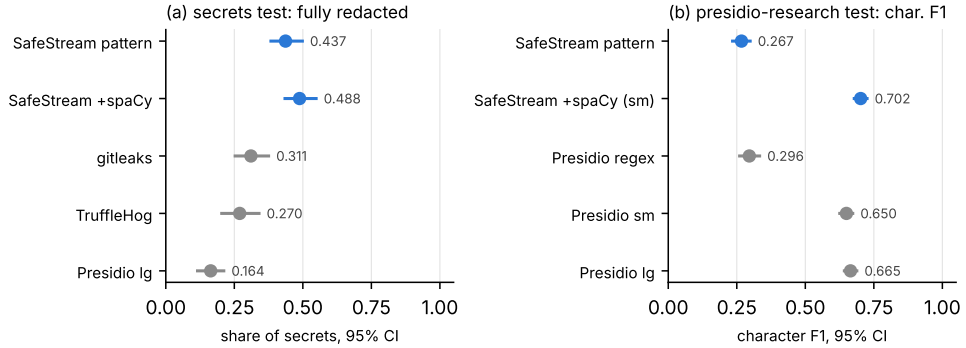


Fig. 5 Test-split accuracy with 95% document-level bootstrap intervals. (a) Share of secrets fully redacted on the secrets test split. (b) Character F1 on the presidio-research test split. SafeStream configurations in blue, baselines in grey.

7.3 RQ3 and RQ4: detection on held-out data

Table 4 and Figure 5 give the test-split results; Table 6 tests the planned differences under clustering.

Credentials (RQ3)

On secrets, SafeStream's pattern tiers fully redacted 0.437 of test secrets, against 0.311 for gitleaks and 0.270 for TruffleHog. With documents resampled independently, the paired differences in fully redacted rate were +0.128 [+0.073, +0.183] and +0.167 [+0.079, +0.250], and in character F1 +0.233 [+0.043, +0.441] and +0.536 [+0.310, +0.685]. The differences in character precision were not significant (+0.005 [−0.063,

Table 4 Test-split results with 95% document-level bootstrap intervals. SafeStream rows are after the entropy fix of Section 7.4, which changed only the secrets results. FR: fully redacted rate; FA: false-alarm document rate.

Corpus / system	Char P	Char R	Char F1	FR	FA
presidio-research (750 documents)					
SafeStream pattern	.995 [.992, .998]	.154 [.133, .176]	.267 [.235, .299]	.115 [.098, .134]	.000
SafeStream +dates	.996 [.993, .998]	.169 [.146, .192]	.290 [.255, .322]	.129 [.110, .150]	.000
SafeStream +spaCy (sm)	.898 [.877, .918]	.576 [.547, .606]	.702 [.679, .725]	.538 [.509, .568]	.079
Presidio regex	.948 [.919, .974]	.175 [.151, .202]	.296 [.260, .333]	.123 [.105, .145]	.000
Presidio sm	.871 [.848, .893]	.518 [.491, .546]	.650 [.626, .673]	.506 [.479, .536]	.048
Presidio lg	.878 [.856, .900]	.535 [.507, .563]	.665 [.642, .688]	.562 [.534, .590]	.000
secrets (313 documents)					
SafeStream pattern	.857 [.705, .946]	.829 [.740, .887]	.842 [.736, .906]	.437 [.384, .498]	.010
SafeStream +spaCy (sm)	.757 [.604, .858]	.845 [.762, .897]	.799 [.683, .870]	.488 [.435, .548]	.340
gitleaks	.837 [.630, .962]	.473 [.299, .679]	.604 [.430, .765]	.311 [.253, .375]	.000
TruffleHog	.803 [.719, .878]	.189 [.117, .308]	.306 [.204, .446]	.270 [.204, .340]	.000
Presidio regex	.282 [.211, .356]	.105 [.063, .177]	.153 [.098, .225]	.126 [.083, .168]	.392
Presidio sm	.251 [.191, .315]	.119 [.072, .197]	.161 [.107, .232]	.160 [.112, .206]	.649
Presidio lg	.308 [.238, .381]	.131 [.082, .214]	.184 [.124, .262]	.164 [.116, .212]	.433

+0.123] and +0.060 [−0.109, +0.176]). Resampling by rule widens every interval (Table 6). The advantage over gitleaks remains significant in both fully redacted rate (two-stage +0.125 [+0.025, +0.209]) and character F1 (+0.226 [+0.011, +0.538]). Against TruffleHog the character-F1 advantage remains (+0.530 [+0.192, +0.742]), but the fully-redacted-rate advantage does not (+0.184 [−0.008, +0.363]); that margin is not robust to dependence between examples of the same rule. SafeStream flagged one of 97 generated non-secrets; neither scanner flagged any. More than half of the test secrets (0.563) were not fully redacted by SafeStream either. The development recall (0.542) exceeded the test recall (0.437) because the test half holds vendor formats for which SafeStream has no pattern. Presidio has no credential recognisers, and adding spaCy raised SafeStream's false-alarm rate to 0.340; on the development split those false alarms were the word `Merge` in commit lines tagged as a person and digest or UUID tokens tagged as persons or organisations.

This comparison has a fairness caveat that we could not resolve with the available runs. The gold labels are Nosey Parker's rule examples, and both scanners apply false-positive filters (allowlists of placeholder-like values in gitleaks, result filtering in TruffleHog) designed to suppress example-like strings. Default configurations reflect how the tools are used, but the corpus may penalise these filters more than real data would. A sensitivity run with the filters disabled was not performed (Section 8.5). SafeStream's own credential patterns were not developed on this corpus: it was added after the split was fixed, and the only change made using it was the suppression rule of Section 7.4, designed on the development half.

Personal data (RQ4)

On presidio-research, the pattern tiers were precise (character precision 0.995) but recalled only 0.154 of sensitive characters, because most gold spans are names, locations and organisations that patterns cannot capture. Against Presidio's pattern recognisers, the paired difference in character recall was −0.021 [−0.041, −0.002] and in precision +0.047 [+0.021, +0.076]; under template clustering the precision advantage remains (+0.046 [+0.011, +0.103]) and the recall deficit is no longer significant. With the same spaCy model, SafeStream +spaCy exceeded Presidio sm in character F1 by +0.051 [+0.027, +0.076] and in fully redacted rate by +0.031 [+0.004, +0.060]. The F1 advantage is robust to template clustering (two-stage +0.052 [+0.013, +0.093]); the fully-redacted-rate advantage is not (+0.033 [−0.014, +0.080]). Against Presidio lg, which uses a larger model, SafeStream +spaCy had higher character F1 with documents resampled independently (+0.036 [+0.011, +0.064]), but not under template clustering (+0.037 [−0.003, +0.080]). The difference in fully redacted rate, −0.025 [−0.053, +0.004], was not significant under any scheme. This corpus is synthetic and template-generated; the comparison shows relative behaviour on that distribution only.

7.4 False alarms on high-entropy non-secrets

Before the fix described in Section 4.1, SafeStream flagged 83 of the 103 generated non-secrets in the development split (false-alarm rate 0.806), all through the entropy tier: every UUID, commit hash, SHA-256 digest and base64 image payload, and none of the minified JavaScript. The suppression rule was designed on the development split. A first version cost one development secret, a UUID-shaped access key assigned to `SAUCE_ACCESS_KEY`, which led to the secret-key veto. With the veto, the development false-alarm rate fell to 0.000 with no change in recall, and the presidio-research and RedactionBench development results were unchanged except in the fourth decimal. On the test split, the false-alarm rate fell from 0.794 [0.714, 0.872] to 0.010 [0.000, 0.035], character precision rose from 0.589 to 0.857, and recall was unchanged (0.829 character recall, 0.437 fully redacted). The test half had been scored once with the pre-fix code before the fix was made; both scores are reported. Because the gold-free documents in both halves come from the same generator, the test improvement is an optimistic estimate of the fix's effect on real non-secrets.

7.5 RQ5: document-level protection and contamination

Table 5 separates the contaminated and uncontaminated results. Rules written after inspecting the benchmark raised test mandatory recall of the pattern tiers by +0.091 [+0.044, +0.163] and character F1 by +0.060 [+0.029, +0.108] relative to the pre-exposure code (paired), and both gains are robust to domain clustering. Because those rules were written while the test documents were visible, this gain is an upper bound on what they contribute to unseen data. One of them, the label-anchored tier, also improved presidio-research, a corpus it was never tuned on (Table 7), which is evidence that it generalises. Development-set selection chose the same configuration for both code versions (spaCy on, entropy on, confidence threshold 0.3).

Table 5 RedactionBench test half (100 documents), with 95% document-level bootstrap intervals. *Exposed*: current code, whose rules were written after inspecting all 200 documents. *Pre-exposure*: the last commit before the benchmark entered the repository. *Dev-selected*: configuration chosen on the development half by a fixed rule (maximum development character F1) and scored on the test half once.

System	Char F1	Mandatory recall	No-leak docs
Pre-exposure, pattern	.518 [.434, .593]	.589 [.476, .696]	.074 [.022, .129]
Pre-exposure, +spaCy	.472 [.395, .547]	.674 [.566, .779]	.137 [.073, .208]
Pre-exposure, dev-selected	.469 [.394, .543]	.676 [.568, .781]	.137 [.073, .208]
Exposed, pattern	.579 [.490, .662]	.682 [.578, .787]	.084 [.032, .144]
Exposed, +spaCy	.512 [.429, .590]	.761 [.657, .860]	.179 [.105, .263]
Exposed, dev-selected	.508 [.427, .588]	.762 [.658, .862]	.179 [.105, .263]
gitleaks	.214 [.130, .314]	.247 [.140, .372]	.021 [.000, .053]
TruffleHog	.030 [.017, .048]	.024 [.013, .040]	.000
Presidio regex	.323 [.250, .393]	.331 [.227, .422]	.011 [.000, .033]
Presidio sm	.336 [.265, .402]	.416 [.326, .500]	.084 [.032, .146]
Presidio lg	.344 [.277, .409]	.420 [.333, .499]	.105 [.052, .174]

Pre-exposure code against the baselines

The pre-exposure code never saw the benchmark, so it gives the uncontaminated comparison. We re-ran it, Presidio, gitleaks and TruffleHog on the test half, saved per-document predictions, and scored all of them on one shared resample matrix (Table 6). With spaCy, its mandatory recall exceeded Presidio lg by +0.262 [+0.091, +0.436] and Presidio sm by +0.265 [+0.093, +0.441]. Both advantages remain significant when domains are resampled (two-stage +0.255 [+0.026, +0.453] and +0.259 [+0.037, +0.454]). Its character F1 exceeded Presidio lg by +0.129 [+0.033, +0.223] and Presidio sm by +0.138 [+0.047, +0.230]. Only the latter is robust: against Presidio lg the two-stage interval is [−0.024, +0.242]. The pattern tiers alone had higher character F1 than Presidio lg (+0.176 [+0.052, +0.289]) and sm (+0.184 [+0.061, +0.297]) at the document level, but neither difference survives domain clustering. Their mandatory-recall advantage over Presidio lg, +0.176 [−0.001, +0.353], was not significant even at the document level. Against the secret scanners every pre-exposure configuration was ahead under all three schemes; for example, the pattern tiers exceeded gitleaks in mandatory recall by +0.342 [+0.263, +0.419] (two-stage +0.335 [+0.191, +0.479]). No-leak rates did not differ significantly from Presidio sm or lg in either configuration (with spaCy against lg: +0.032 [−0.042, +0.102]).

No-leak documents

The no-leak document rate was low for every system: at most 0.179 [0.105, 0.263] for SafeStream (+spaCy, exposed code), 0.137 for the pre-exposure code with spaCy, 0.084 for the pattern tiers alone, and at most 0.105 for Presidio. In other words, for at least 82% of RedactionBench test documents, every configuration we evaluated left at least one must-redact character in the clear. This metric is deliberately strict. Mandatory recall measures the share of must-redact characters hidden; the no-leak rate requires all of them in a document to be hidden, so a single missed character fails the document. RedactionBench documents are dense: the test half averages about 256 gold spans per document (Table 2), of which 84 on average (median 25) are mandatory. As an illustration only, if a document had n independent must-redact units each

Table 6 Planned paired comparisons under three bootstrap schemes: median paired difference and 95% interval (2,000 resamples, seed 0, one shared resample matrix per corpus). † marks an interval that includes zero. Pre-exposure SafeStream on RedactionBench is compared with every baseline because it never saw the benchmark.

Comparison	Metric	Document	Cluster	Two-stage
secrets: clusters are 96 rules and 5 negative templates				
Pattern − gitleaks	FR	+.128 [+.073, +.183]	+.123 [+.033, +.206]	+.125 [+.025, +.209]
Pattern − gitleaks	Char F1	+.233 [+.043, +.441]	+.215 [+.013, +.527]	+.226 [+.011, +.538]
Pattern − TruffleHog	FR	+.167 [+.079, +.250]	+.178† [−.010, +.350]	+.184† [−.008, +.363]
Pattern − TruffleHog	Char F1	+.536 [+.310, +.685]	+.529 [+.192, +.740]	+.530 [+.192, +.742]
presidio-research: clusters are 200 templates				
+spaCy − Presidio sm	Char F1	+.051 [+.027, +.076]	+.052 [+.013, +.092]	+.052 [+.013, +.093]
+spaCy − Presidio sm	FR	+.031 [+.004, +.060]	+.032† [−.015, +.077]	+.033† [−.014, +.080]
+spaCy − Presidio lg	Char F1	+.036 [+.011, +.064]	+.036† [−.000, +.075]	+.037† [−.003, +.080]
RedactionBench, pre-exposure SafeStream: clusters are 11 domains				
+spaCy − Presidio lg	Mand. recall	+.262 [+.091, +.436]	+.252 [+.085, +.348]	+.255 [+.026, +.453]
+spaCy − Presidio sm	Mand. recall	+.265 [+.093, +.441]	+.256 [+.104, +.345]	+.259 [+.037, +.454]
+spaCy − Presidio lg	Char F1	+.129 [+.033, +.223]	+.124 [+.006, +.221]	+.123† [−.024, +.242]
+spaCy − Presidio sm	Char F1	+.138 [+.047, +.230]	+.132 [+.031, +.219]	+.132 [+.002, +.242]
+spaCy − Presidio lg	No-leak	+.032† [−.042, +.102]	+.036† [−.096, +.157]	+.031† [−.110, +.169]
Pattern − Presidio lg	Char F1	+.176 [+.052, +.289]	+.167† [−.051, +.298]	+.169† [−.068, +.319]
Pattern − Presidio lg	Mand. recall	+.176† [−.001, +.353]	+.168† [−.252, +.266]	+.164† [−.239, +.373]
Pattern − gitleaks	Mand. recall	+.342 [+.263, +.419]	+.340 [+.217, +.468]	+.335 [+.191, +.479]
Pattern − TruffleHog	Mand. recall	+.571 [+.456, +.666]	+.560 [+.304, +.642]	+.563 [+.315, +.677]
+spaCy − gitleaks	Mand. recall	+.427 [+.350, +.510]	+.433 [+.320, +.561]	+.439 [+.297, +.589]
+spaCy − TruffleHog	Mand. recall	+.656 [+.548, +.749]	+.651 [+.591, +.713]	+.662 [+.526, +.751]
RedactionBench, contamination: exposed minus pre-exposure pattern tiers				
Exposed − pre-exposure	Mand. recall	+.091 [+.044, +.163]	+.094 [+.064, +.159]	+.089 [+.043, +.198]

detected with probability p, it would be leak-free with probability p^n, which for n in the hundreds is small even for p close to one. Real misses are correlated (whole entity types missing in a domain), so this is not a model of the data, but it shows why per-character recall in the 0.68–0.76 range is compatible with almost no fully protected documents.

Where the leaks are. We re-ran five systems of Table 5 (exposed pattern tiers, exposed +spaCy, gitleaks, TruffleHog and Presidio lg) on the test half at the commit that produced those rows, reproducing its mandatory recall and no-leak rates exactly, and kept the per-document predictions. The 95 test documents with mandatory gold contain 8,002 mandatory spans. Three observations follow; they are descriptive, from

Table 7 Ablation on the test splits (pure-Python backend, identical output to the Rust core). Paired differences against the full pattern pipeline, with 95% document-level intervals.

Change	secrets	presidio-research	RedactionBench
Character F1, full pattern pipeline minus variant			
Remove entropy tier	+.455 [.242, .641]	.000	+.156 [.104, .224]
Remove label-anchored tier	.000	+.030 [.019, .044]	+.004 [.002, .006]
Remove contextual detections	.000	.000	.000
Union rule versus greedy resolution (with spaCy), union minus greedy			
Character recall	+.006 [.000, .017]	+.005 [.002, .008]	+.009 [.004, .016]
FR or mandatory recall[1]	+.013 [.000, .032]	+.001 [.000, .002]	+.012 [.004, .022]

[1]Fully redacted rate (FR) for secrets and presidio-research; mandatory recall for RedactionBench.

one corpus that had been inspected during development, and carry no confidence intervals.

- *Leaks are whole misses, not boundary errors.* With spaCy, 6,441 mandatory spans were fully redacted, 64 partially and 1,497 not at all; of the 17,122 leaked mandatory characters, 826 (4.8%) lie in partially redacted spans and the rest in spans with no detection. For the pattern tiers alone the share is 3.7% (854 of 22,778). Presidio lg leaks more through partial coverage: 7,354 of its 41,589 leaked characters (17.7%).
- *Few documents are near misses; leaks are concentrated.* With spaCy, 17 documents leaked nothing, 5 more leaked exactly one mandatory span and 7 more leaked two, so fixing one or two spans per document would raise the no-leak rate to at most 29 of 95 (0.305). The median document leaked 6 mandatory spans (interquartile range 2–16), a median of 23% of its mandatory spans, and the 9 documents with the most leaked characters held 67% of them. For the pattern tiers alone the median document leaked 11 spans (52%), and only 2 documents were within two spans of no leak.
- *What is missed.* The corpus carries no entity types, so we grouped spans by a coarse shape heuristic. Capitalised words (largely names) leaked at a rate of 0.90 without a named-entity model and 0.27 with spaCy, which accounts for most of the gap between the two configurations. Spans containing `@` leaked at 0.95 (36 of 38) with spaCy; inspection of a few of them, after all scores were final, showed passwords inside connection strings or quoted configuration values that the e-mail pattern split or that no tier matched. No code was changed in response, and these spans should not be read as a test-set error analysis used for tuning.

Even an oracle that took the union of all five systems' redactions would leave 69 of 95 documents leaking (no-leak rate 0.274), so the low document-level rate is a property of the corpus and of current detectors in general, not only of SafeStream. The low rate is not caused by streaming: by Theorem 1 the streamed output equals the whole-document output, so these leaks are detection misses that a whole-document run would also make.

7.6 Ablation

Table 7 isolates the components. The entropy tier carried most of the credential recall: without it, character F1 on secrets fell from 0.842 to 0.383. The label-anchored tier

mattered on presidio-research, where several identifiers carry labels. The contextual scorer's own detections changed no result on these corpora; its confidence adjustment was kept in every variant. Alone, the deterministic tier reached a character F1 of 0.380 on secrets and 0.237 on presidio-research, and the entropy tier alone 0.822 on secrets. The union rule raised recall by small and mostly significant amounts at no significant cost in F1 on secrets (+0.000 [−0.003, +0.005]) and RedactionBench (+0.003 [−0.002, +0.008]); on presidio-research its F1 difference was +0.004 [+0.001, +0.007]. The ablation was not planned and its intervals are document-level only.

8 Discussion

8.1 What the results establish, and what they do not

Established under stated assumptions: chunked and parallel processing introduce no detection difference relative to the whole-document pipeline, for detectors that respect their declared bounds (Section 5). *Measured:* flat working memory across a 256-fold range of input sizes on one input shape; near-linear scaling to four cores; on held-out splits, higher credential coverage than two scanners in default configurations on a corpus of rule examples (robust to rule clustering against gitleaks, and in character F1 against TruffleHog); and higher character F1 than Presidio with the same spaCy model on a synthetic corpus, robust to template clustering. *Not established:* effectiveness on real-world artefacts; document-level protection adequate for public release; any property of the unevaluated transformer tier; robustness to adversarial obfuscation.

8.2 When is the approach useful despite a low no-leak rate?

Bounded memory and equivalence do not make redaction complete; they remove one specific failure mode, the leak introduced by chunking itself, and make large inputs processable at all. The value of that depends on the detection coverage for the values that matter in a given deployment. The evidence supports three uses and argues against a fourth.

1. *Defence in depth for structured secrets in machine-generated text.* Where the sensitive values are credentials and structured identifiers with recognisable shapes (logs, configuration dumps, continuous-integration output), the pattern tiers are where coverage is highest in our data, and the guarantee ensures that splitting a multi-gigabyte log does not make coverage worse. Even here more than half of the test secrets were not fully redacted, so the tool reduces exposure rather than eliminating it.
2. *Pre-filtering before human review or controlled sharing.* When output is reviewed, or shared only with parties already bound by agreements, a high-precision filter that reliably processes large inputs reduces the volume of exposed values without being the last line of defence.
3. *Inputs that cannot otherwise be processed.* Whole-document redaction needed about 28 times the input size in memory in our measurements, which rules it out for multi-gigabyte files on ordinary hosts.

4. *Not supported: sole control for public release of free text.* With no-leak rates below 0.18 on RedactionBench and low free-text recall without a named-entity model, the results do not justify releasing redacted free-text documents publicly on the strength of this system alone.

A deployment should set its own acceptance criteria before relying on the system, for example: per-category recall measured on a representative, authorised sample of the target artefacts; zero tolerated leaks for specified credential types on that sample; a review step for document classes with free text; and periodic re-measurement as formats change. None of these was measured here.

8.3 Threats to validity

External validity. presidio-research is synthetic and template-generated; the secrets positives are rule-author examples and the negatives are generated from five templates; RedactionBench is the only corpus with real-world seeding, and it is contaminated. The corpora are English-only. No result here establishes performance on real logs or exports. *Construct validity.* The secrets gold spans are Nosey Parker's capture groups, so credentials adjacent to a labelled secret (for example a client identifier next to its secret) count as false positives; RedactionBench's contextual labels have low human agreement according to its authors [33]. *Internal validity.* The secrets test half was scored before and after one fix (Section 7.4); RedactionBench was inspected during development. *Statistical validity.* The planned comparisons were tested under document, cluster and two-stage resampling, and Table 6 shows which differences depend on the independence assumption. Eleven RedactionBench domains are few clusters for a bootstrap, so those intervals are approximate. The ablation intervals are document-level only, and no multiplicity correction was applied. *Comparison fairness.* The scanners were run with default filters on a corpus of examples those filters target. *Environment.* All runs used one virtual machine.

8.4 The unevaluated neural tier

A transformer tier (a DistilBERT [20] model exported to ONNX) and its gate are implemented but contribute no result in this paper; its models and the corpora for evaluating it are hosted on a service the evaluation environment could not reach. Evaluating it requires: (i) data: the ai4privacy test split [36] under its licence and the Text Anonymization Benchmark [17], excluding CoNLL-2003 from claims because the bundled model was trained on it [18]; (ii) baselines: Presidio with transformer models and a generalist span model [21]; (iii) metrics: the same character, fully-redacted and no-leak metrics, plus throughput and peak memory; (iv) an ablation of the gate (fraction of characters reaching the model and recall lost); (v) an empirical equivalence check, comparing streamed and whole-document output with the model enabled, because the theorems do not cover a model whose output depends on its whole input window; (vi) pinned model files, hashes and split manifests.

8.5 Evaluation plan for real-world evidence

Ranked by expected value per unit of effort:

1. *Scanner sensitivity analysis:* gitleaks and TruffleHog with their false-positive filters disabled, reported alongside the defaults.
2. *Real credential-bearing text with clear provenance:* public continuous-integration logs or issue attachments from repositories whose licences permit research use, or logs from an organisation that authorises the study. Safeguards: never test discovered credentials against live services; store raw text encrypted with access limited to annotators; report discovered live secrets to their owners through responsible disclosure; publish only annotations, hashes and aggregate results, not secret values; obtain ethics review where required; double annotation with agreement reported.
3. *The neural-tier evaluation* of Section 8.4.
4. *A second machine* for throughput and memory, to separate implementation effects from the virtual machine.

9 Conclusion

Chunked redaction need not trade correctness for scale. With detector-declared bounds on span length and context, an overlap that covers every detector's span plus its right context makes the streamed result equal to the whole-document result, a seam cross-check makes the parallel result equal to the sequential one, and the working buffer does not grow with the input. In our measurements memory stayed flat from 8 MiB to 2 GiB and throughput scaled 3.9-fold on four cores. These properties remove chunking as a source of leaks; they do not make detection complete. On held-out data, the pattern tiers covered more credentials than gitleaks and TruffleHog in default configurations, and with a spaCy model SafeStream was more accurate than Presidio with the same model on a synthetic corpus. Some of these margins do not survive cluster-robust resampling, fewer than one RedactionBench document in five was fully protected by any configuration, and two of the three corpora are synthetic. The system is therefore suited to reducing exposure in large machine-generated artefacts as one layer among several, not to certifying documents for public release. The next steps are evaluation on authorised real-world text, the neural tier, and a scanner sensitivity analysis (Section 8.5).

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- **Competing interests.** [MISSING: The authors have no relevant financial or non-financial interests to disclose. – confirm or replace]
- **Ethics approval and consent to participate.** Not applicable: no human participants were involved. All corpora are public or generated. The secrets corpus consists of example strings published in an open-source rule set; no credential was

tested against a live service (TruffleHog ran without verification). The ai4privacy corpus was not used.

- **Consent for publication.** Not applicable.
- **Data availability.** presidio-research and Nosey Parker are public; RedactionBench is available from its authors under CC BY 4.0. Generated non-secrets are produced by the released code with a fixed seed. Split manifests, raw results and per-document predictions (character offsets only, no corpus text) are in the repository under `research/results/`. [MISSING: archive a tagged release (e.g. on Zenodo) and give its DOI here].
- **Code availability.** Source code and the evaluation harness, including the cluster-bootstrap script `benchmarks/cluster_bootstrap.py`, are available at https://github.com/MounishSenisetty/SafeStream-Redactor.
- **Author contributions.** [MISSING: each author's contribution (conceptualisation, software, validation, formal analysis, investigation, writing)]
- **Use of generative AI.** Claude (Anthropic) was used to implement evaluation scripts and to draft the manuscript text. The authors reviewed and edited the content and take full responsibility for the publication.

Appendix A Proofs of Theorems 2 and 3 and implementation mapping

Throughout, positions are absolute offsets in T, the conditions of Theorem 1 hold unless stated otherwise, and $L_{\max} = \max_d L_d \le w$.

A.1 A streaming engine started mid-text

The parallel engine runs Algorithm 1 on a substring of T with a read-only left context. Lemma 1 covers that case.

Lemma 1 *Let $X = T[x_0{:}x_1]$ and let the engine start with $P = T[x_0{:}x_0 + h]$ as already-emitted context and stream $T[x_0 + h : x_1]$, where $h = w$, or $h < w$ and $x_0 = 0$. Then it emits exactly the spans of $\Pi(X)$ that start at or after $x_0 + h$.*

Proof The proof of Theorem 1 uses only two properties of P: it holds the $\min(w, p - x_0)$ characters preceding B, and the window ends at the end of the text being processed. Both hold here with X in place of T. Spans of the first window that start inside P are dropped, as in every later pass. □

Lemma 2 (Transfer to the full text) *With $X = T[x_0{:}x_1]$, let s be a span of detector d that belongs to $\Pi(X)$ or to $\Pi(T)$, with $i - C_{\text{left},d} \ge x_0$ or $x_0 = 0$, and $j + C_{\text{right},d} \le x_1$ or $x_1 = |T|$. Then $s \in \Pi(X)$ if and only if $s \in \Pi(T)$.*

Proof Apply A2 with $a = x_0$ and $b = x_1$ in T; in X these positions are its two edges. This gives $\delta = -x_0$ in the notation of A2, and an edge of X that coincides with an edge of T satisfies the edge clause. □

A.2 Proof of Theorem 2

Let the split points be $0 = a_0 < a_1 < \cdots < a_k = |T|$, with $a_{r+1} - a_r \geq 4w$; `plan_ranges` enforces this, and coarsens the split or falls back to one range otherwise. Worker r processes $X_r = T[x_0^r : x_1^r]$ with $x_0^r = \max(0, a_r - 2w)$ and $x_1^r = \min(|T|, a_{r+1} + w)$. Its first w characters, $T[a_r - 2w : a_r - w]$, are read-only context, and streaming starts at $a_r - w$; worker 0 has no context and starts at 0. By Lemma 1, worker r obtains D_r, the spans of $\Pi(X_r)$ that start at or after $a_r - w$.

A *straddler* of a seam a is a span with $i < a < j$. Worker r reports the straddler of a_r (its start seam) and the straddler of a_{r+1} (its end seam) found in D_r, or reports that none exists. It emits $T[\ell_r : u_r]$, where ℓ_r is the end of its start-seam straddler or a_r, and u_r is the end of its end-seam straddler or a_{r+1}. Together with that text it emits the spans of D_r contained in $[\ell_r, u_r)$.

Step 1: in the window that matters, worker r sees $\Pi(T)$. Let s be a span of detector d with $a_r - w \leq i < a_{r+1}$ that belongs to D_r or to $\Pi(T)$. On the left, $i - C_{\text{left},d} \geq a_r - 2w \geq x_0^r$, because $C_{\text{left},d} \leq w$. On the right, $j \leq i + L_d < a_{r+1} + L_d \leq a_{r+1} + w - C_{\text{right},d}$, so $j + C_{\text{right},d} \leq x_1^r$, or $x_1^r = |T|$. By Lemma 2, $s \in D_r \Leftrightarrow s \in \Pi(T)$.

Step 2: both neighbours report the same straddler. A straddler of a_r starts after $a_r - L_d \geq a_r - w$, and a straddler of a_{r+1} starts before a_{r+1}, so both lie in the range of Step 1. Spans that start before $a_r - w$ are dropped as context. Such a span ends at most $L_{\max} \leq w$ characters after its start, that is, at or before a_r, so it cannot cross a_r. Each straddler found by worker r is therefore the unique straddler of $\Pi(T)$ at that seam, if one exists (uniqueness follows from disjointness). Worker $r + 1$ applies Step 1 to its own start seam a_{r+1} and obtains the same span. The reports agree, so the valve passes, and $u_r = \ell_{r+1}$.

Step 3: each part is the sequential redaction of its slice. Every span emitted by worker r lies in $[\ell_r, u_r)$. It starts at or after $\ell_r \geq a_r$, and before a_{r+1}: if there is an end-seam straddler, any span starting at or after a_{r+1} and ending by u_r would overlap it; otherwise $u_r = a_{r+1}$. By Step 1 the emitted spans are exactly the spans of $\Pi(T)$ inside $[\ell_r, u_r)$. No cut point ℓ_r lies strictly inside a span of $\Pi(T)$, because it is either a seam with no straddler or the end of a span of $\Pi(T)$, and $\Pi(T)$ is disjoint. The slices $[\ell_r, u_r)$ therefore partition T without cutting any span. Redaction acts on each span separately (Definition 1), so the concatenated parts equal the redaction of T under $\Pi(T)$. By Theorem 1 that is the sequential output.

Step 4: timing of the cuts. A worker writes text only once the cuts that can affect it are fixed. It decides ℓ_r once its stream has passed $a_r + w$. By then every straddler of a_r, which ends before $a_r + L_{\max}$, has been emitted by the streaming engine. It flushes only text below the current position minus $2w$. The stream never extends beyond $a_{r+1} + w$, so this text lies below $a_{r+1} - w$, and the end cut $u_r \geq a_{r+1}$ cannot affect it. This proves (i).

Proof of (ii). This part uses no assumption on the detectors. For each seam, the coordinator compares the left worker's end report with the right worker's start report

and redacts the whole file sequentially on any mismatch. If all reports match, each u_r equals ℓ_{r+1}, because both are computed from the same reported straddler end. The written parts then cover $[0, |T|)$ as consecutive, non-overlapping slices, and every character of T is emitted exactly once, redacted or not. A non-splittable text encoding, a single worker or a small file also take the sequential path. When the conditions of Theorem 1 fail, agreeing parts can still contain different detections from the sequential output; this is why an overlap below $w^\star$ is refused by default. □

A.3 Proof of Theorem 3

After an emission. The retained buffer is $B[e{:}]$. If no span straddles the planned boundary, $|B| - e = w$. Otherwise $e = i^* > |B| - w - L_d$, so $|B| - e < w + L_{\max}$. In both cases the retained buffer holds fewer than $w + L_{\max} \le w + m$ characters.

Before a detection pass. Chunks are appended until $|B| \ge w + m$. Before the last of them, $|B| < w + m$, and that chunk adds at most c characters, so a pass starts with $|B| < w + m + c$. The proof of Theorem 1 shows that the pinned branch, which would let B grow further, is never taken when $m \ge w \ge L_{\max}$. The final buffer obeys the same bound. The window $P \cdot B$ adds $|P| \le w$, which gives $2w+m+c$. The detections of a window are pairwise disjoint and lie inside it, so their number is at most its length.

Parallel workers. Each worker runs one streaming engine and holds back emitted segments until they lie $2w$ behind its stream position, together with the detections of that suffix. This adds at most $2w$ characters plus one segment of at most $w + m + c$ characters. The bound does not depend on $|T|$, and k workers need at most k times as much. □

A.4 Implementation mapping

Table A1 Proof steps and the code that discharges them (module paths under `src/safestream_redactor/`).

Proof step	Code and tests
Declared $(L_d, C_{\text{left},d}, C_{\text{right},d})$ and $\max(L_d + C_{\text{right},d}, C_{\text{left},d})$	`detectors/base.py`: `StreamingRequirements`, `required_overlap`
$w^\star$ (Eq. 1); refusal below it	`engine.required_overlap`; `Redactor.min_safe_overlap`
Window $P \cdot B$, spans inside P dropped	`engine.stream_windows`, `window_detect`
Planned boundary and retreat	`emit_upto = len(buffer) - overlap`; `_retreat_boundary`
Span longer than w fails closed	`_check_overlong` (`strict_spans`)
Coalescing to m; pinned branch	`min_emit`, `need` in `stream_windows`
Disjoint detection set	`entities.resolve_overlaps`
Span-local rewrite	`policy.RedactionPolicy.apply`, `render`
Ranges of at least $4w$	`parallel.plan_ranges`
Context and padding of X_r	`parallel._redact_range`, `_read_chars_before`, `_read_chars_after`
Straddler reports; cuts ℓ_r, u_r	`parallel._find_straddler`; `start_cut`, `end_cut`
Holdback of $2w$	`flush(cursor - 2 * ov)` in `_redact_range`
Seam valve and fallback	`redact_file_parallel`: `left.end_seam != right.start_seam`
Empirical corroboration	`tests/test_streaming.py`, `test_properties.py`, `test_parallel.py`, `test_streaming_contract.py`, `test_formal_constants.py`, `test_memory.py`

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